CMPE 401 Instructor-defined Project 1

Advanced Object Detection and Comparative Study using YOLOv11 (or YOLOv26)

1. Project Title

Design, Optimization, and Comparative Evaluation of Modern YOLO Models for Real-World Object Detection

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2. Background & Motivation

Real-time object detection in computer vision plays a central role in:

- Autonomous driving
- Robotics
- Smart surveillance
- Industrial inspection
- Smart cities

The YOLO (You Only Look Once) family, maintained by Ultralytics, is one of the most widely used real-time detection frameworks in industry.

Modern versions such as YOLOv11 (and recent updates such as YOLOv26) aim to improve:

- Speed–accuracy trade-off
- Small-object detection capability
- Model efficiency
- Training stability

This project focuses on building a complete experimental pipeline using YOLOv11 (or YOLOv26), performing structured experiments, systematic analysis, and multi-version comparisons.

https://docs.ultralytics.com/models/yolo11/
https://docs.ultralytics.com/models/yolo26/
https://github.com/ultralytics/ultralytics

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3. Project Objectives

Students will:

1. Understand the YOLO detection pipeline
2. Fine-tune a modern YOLO model
3. Design and execute structured experiments
4. Analyze training and validation loss behavior
5. Diagnose overfitting and underfitting
6. Perform iterative model improvements
7. Compare multiple YOLO versions quantitatively
8. Present reproducible research results via Github

4. Dataset

VisDrone (Recommended)

https://github.com/VisDrone/VisDrone-Dataset

You can choose the first image detection subset.

Task 1: Object Detection in Images

VisDrone-DET dataset

- trainset (1.44 GB): BaiduYun | GoogleDrive
- valset (0.07 GB): BaiduYun | GoogleDrive
- testset-dev (0.28 GB): BaiduYun | GoogleDrive (GT avalialbe)
- testset-challenge (0.28 GB): BaiduYun | GoogleDrive

VisDrone-DET toolkit:

- Matlab beta

Features:

- Small objects
- Dense scenes
- Real-world complexity
- Suitable for small-object analysis

Please challenge your final model using the testset-challenge. This will serve as a competition. Projects that demonstrate top-performing model in the class on the challenge test set, supported by rigorous experimental analysis and clear justification of design choices, will be considered as exceeding expectations in grade.

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5. Required Technical Project Tasks

Part I – Baseline Model

- **Train YOLOv11 or YOLOv26 as baseline model**

Record Result:

- Training loss curve
- Validation loss curve

- mAP
- Precision
- Recall
- etc

Part II – Loss Curve and Fitting Analysis

Students should:

- Plot training vs validation loss
- Identify convergence behavior
- Discuss overfitting
- Discuss underfitting
- Explain causes

Discussion must reference:

- Dataset size
- Model capacity

Part III – Structured Experimental Design

Students must design at least ONE ROUND controlled experiments.

Each experiment must:

- Clearly state experimental settings
- Report quantitative results
- Provide analysis

Suggested variables:

- Model size (n / s / m)
- Image resolution
- Data augmentation strategy
- Learning rate scheduling
- Batch size
- Training epochs
- etc

Part IV – Iterative Model Improvement

Under Part III, students must perform at least ONE improvement cycles grounded in deep learning principles.

Each improvement must follow:

Baseline -> Experimental Settings -> Controlled Modification -> Evaluation -> Analysis -> Conclusion

Examples:

- Regularization
- Normalization
- Weight Initialization
- Learning rate schedule
- Overfitting control
- etc

Each iteration must include:

- Quantitative comparison table
- Performance discussion
- Justification of change

Part V – Multi-Version YOLO Comparison

Students must compare YOLOv11 (or YOLOv26) with at least THREE other YOLO versions.
Possible comparison models:

- YOLOv8
- YOLOv5
- YOLOv9
- YOLOv10
- etc

Comparison must include:

- mAP
- Precision
- Recall
- Model size
- Training time
- Inference speed
- etc, other performance, like confusion matrix

A structured comparison table is required.

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6. Project Summary

This project requires students to design, train, analyze, and systematically evaluate a modern object detection system using YOLOv11 (or YOLOv26). The focus is not only on model performance, but on understanding training dynamics, experimental design, and comparative analysis.
Students must:

- Build a complete detection pipeline
- Analyze training and validation loss behavior
- Diagnose overfitting and underfitting
- Conduct controlled experiments
- Perform theory-driven improvements
- Compare multiple YOLO versions quantitatively
- Present reproducible results via GitHub

The goal is to develop both engineering implementation skills and a rigorous understanding of modern deep learning systems through structured experimentation and critical analysis.